

Theme

Comparative Analysis of Proposer Advantage in Stable Matching: A Job-Applicant Market Application

Goals, data, methods, models

Goals: The primary goal of this project is to analyze the structural differences, trade-offs, and fairness outcomes between different matching mechanisms in a two-sided decision market. We aim to quantify the mathematical proposer advantage by directly comparing the outcomes of an Applicant-proposing system versus an Employer-proposing system. As an extension we may also explore how these dynamics shift when introducing capacity constraints (many-to-one matching) or ties in the preference lists.

Data: We will use a publicly available tabular dataset (sourced from Kaggle or GitHub) containing Job Postings (required skills, industry) and Applicant Resumes (candidate skills, education). To ensure a computationally manageable bipartite graph, we will work with a controlled subset of this data.

Methods & Models:

- **Preference Generation:** We will use basic similarity metrics (e.g., cosine similarity of text features/skills) to generate strict, ranked preference lists for both sides of the market (Applicants ranking Jobs, and Jobs ranking Applicants).
- **Models:** The problem will be modeled as a Bipartite Graph, utilizing the **Gale-Shapley algorithm** to find stable matchings.
- **Comparative Analysis:** We will execute both the Applicant-optimal and Employer-optimal variants of the algorithm. We will analyze the resulting graphs to verify stability (the absence of blocking pairs) and statistically compare the average rank satisfaction across both sides to measure the magnitude of the proposer advantage.

Group members

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